

Numerical Simulation of the Armington Model

International Trade

Farid Farrokhi

Introduction

- GE models of trade are commonly used in many applications:
 - Impacts of U.S. tariffs on prices and welfare in the U.S. and elsewhere.
 - ... implications of regional trade agreements such as NAFTA.
 - ... implications of Chinese manufacturing productivity growth.
 - ... and many more.
- Numerical simulations are often required to answer these questions quantitatively.
- In these slides, we focus on the Armington model to overview some numerical methods. However, the methods which we discuss are applicable to more advanced models.

Recap: Armington Model

- Price index:

$$P_j = \left[\sum_{\ell=1}^N b_{\ell j} d_{\ell j}^{1-\sigma} (w_{\ell}/a_{\ell})^{1-\sigma} \right]^{\frac{1}{1-\sigma}} \quad (1)$$

- Trade shares:

$$\pi_{ij} = \frac{b_{ij} d_{ij}^{1-\sigma} (w_i/a_i)^{1-\sigma}}{P_j^{1-\sigma}} \quad (2)$$

- Market clearing condition:

$$w_i L_i = \sum_{j=1}^N \pi_{ij} w_j L_j \quad (3)$$

- Welfare:

$$U_i = \frac{w_i L_i}{P_i} \quad (4)$$

- Eqs (1)-(4) take $\{L_i, a_i, b_{ij}, d_{ij}, \sigma\}$ as given to predict $\{P_i, \pi_{ij}, w_i, U_i\}$.

The GE System

- Recast the entire system of GE onto the following equation:

$$\underbrace{w_i L_i}_{S_i(\mathbf{w})} = \sum_{j=1}^N \underbrace{\frac{b_{ij} d_{ij}^{1-\sigma} (w_i/a_i)^{1-\sigma}}{\sum_{\ell=1}^N b_{\ell j} d_{\ell j}^{1-\sigma} (w_{\ell}/a_{\ell})^{1-\sigma}}}_{D_i(\mathbf{w})} w_j L_j, \quad \text{for all } i = 1, \dots, N. \quad (5)$$

where $S_i(\mathbf{w})$ is labor supply and $D_i(\mathbf{w})$ is labor demand for each country i .

- We seek to find the vector of wage rates, $\mathbf{w} = [w_i]_{i=1}^N$, that solves eq (5), given $\sigma, L_i, a_i, b_{ij}, d_{ij} \forall i, j$.
- We overview two methods:
 - (1) model-specific root-finding algorithms
 - (2) generic root-finding Newton's methods.

Method #1: A model-specific algorithm

- Set tolerance, e.g., at 10^{-9} . Guess $w_i^{(1)} \forall i$ for iteration $t = 1$.
- At iteration $t \geq 1$,
 1. Normalize all wage rates w.r.t. the numeraire (e.g., labor in country i_0): Replace $w_i^{(t)}$ with $w_i^{(t)} / w_{i_0}^{(t)}$.
 2. Calculate labor supply and demand for all i ,

$$S_i^{(t)} = w_i^{(t)} L_i, \quad D_i^{(t)} = \sum_{j=1}^N \frac{b_{ij} d_{ij}^{1-\sigma} \left(w_i^{(t)} / a_i \right)^{1-\sigma}}{\sum_{\ell=1}^N b_{\ell j} d_{\ell j}^{1-\sigma} \left(w_{\ell}^{(t)} / a_{\ell} \right)^{1-\sigma}} w_j^{(t)} L_j$$

3. Calculate error $e = \max_i \left| D_i^{(t)} / S_i^{(t)} - 1 \right|$. If e is smaller than tolerance $= 10^{-9}$, convergence is achieved: Stop. Otherwise, update the current guess of wage, $w_i^{(t)}$, to a new one, $w_i^{(t+1)}$, for all i

$$(UR-1) \quad w_i^{(t+1)} = \left(D_i^{(t)} / S_i^{(t)} \right) \times w_i^{(t)}$$

and, take $w_i^{(t+1)} \forall i$ to iteration $t+1$.

Comments on Method #1

- Wages can be solved only up to a choice of numeraire.
 - We chose labor in a country. Another common choice is to normalize the world GDP, $\sum_i w_i L_i$.
- For iterations to converge, sometimes we need to dampen the updated wages: replace $w_i^{(t+1)}$ from (UR-1) with $(1 - \mu^{(t)})w_i^{(t)} + \mu^{(t)}w_i^{(t+1)}$, that is,

$$(UR-2) \quad w_i^{(t+1)} = (1 - \mu^{(t)})w_i^{(t)} + \mu^{(t)} \overbrace{\left(D_i^{(t)} / S_i^{(t)} \right) w_i^{(t)}}^{(UR-1)}$$

- where $\mu^{(t)} \in (0, 1)$ is a dampening parameter
 - A lower $\mu^{(t)}$ raises the likelihood of convergence but it may reduce its speed.
- The algorithm is silent about the choice of dampening parameter. In practice, we may experiment with it. But later we will have more to say.

Method #1: A model-specific algorithm

- The updating rule in Method #1 relies on the insight that a greater demand for country i 's labor relative to its supply bids up the wage rate w_i there, so that the equilibrium is given a chance to be restored.
- More generally, the updating rule could be written as:

$$(UR-3) \quad w_i^{(t+1)} = g\left(D_i^{(t)} / S_i^{(t)}\right) \times w_i^{(t)}$$

where $g(\cdot)$ is an increasing function, satisfying $g(1) = 1$.

- We later show that updating rules 2 and 3 approximate standard root-finding Newton's methods, but let us first review Newton's methods.

Newton's method

- Consider the problem of finding $f(x) = 0$ for a real-valued continuous and differentiable function, $f(x) : \mathbb{R} \rightarrow \mathbb{R}$.
- Starting with a guess $x^{(t)}$, we can linearize f using its slope at $x^{(t)}$,

$$f\left(x^{(t+1)}\right) \approx f\left(x^{(t)}\right) + f'\left(x^{(t)}\right)\left(x^{(t+1)} - x^{(t)}\right)$$

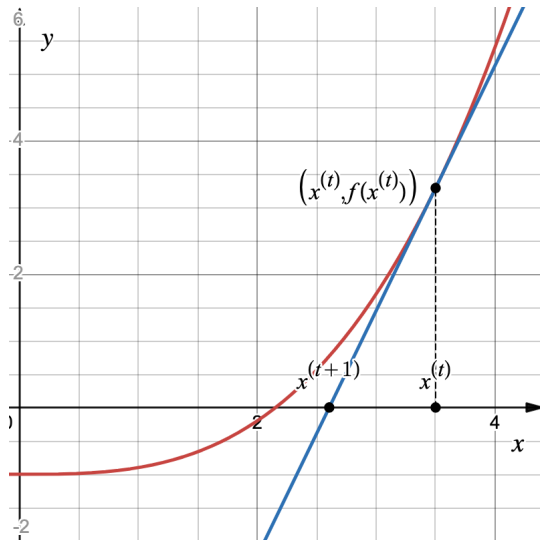
- This line intersects with the x-axis at $f\left(x^{(t+1)}\right) = 0$, where:

$$x^{(t+1)} = x^{(t)} - \frac{f\left(x^{(t)}\right)}{f'\left(x^{(t)}\right)}$$

- The above delivers the updating rule in iteration t of the Newton's method.

Newton's method

- Graphical representation of the Newton's method



Note: In this example, $f(x) = -1 + 0.1x^3$, $x^{(t)} = 3.5$, and $x^{(t+1)} = 3.5 - \frac{-1+0.1 \cdot 3.5^3}{0.3 \times 3.5^2} \approx 2.6$.

Newton's method

- Now, consider the N -dimensional problem,

$$\mathbf{f}(\mathbf{x}) = 0, \quad \mathbf{f} : \mathbb{R}^N \rightarrow \mathbb{R}^N$$

where $\mathbf{x} = [x_i]_{i=1}^N$ and $\mathbf{f}(\mathbf{x}) = [f_i(\mathbf{x})]_{i=1}^N$.

- Similar procedure gives the following updating rule,

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)} - \left[\nabla \mathbf{f}(\mathbf{x}^{(t)}) \right]^{-1} \mathbf{f}(\mathbf{x}^{(t)})$$

$$\begin{bmatrix} x_1^{(t+1)} \\ \vdots \\ x_N^{(t+1)} \end{bmatrix} = \begin{bmatrix} x_1^{(t)} \\ \vdots \\ x_N^{(t)} \end{bmatrix} - \begin{bmatrix} \frac{\partial f_1(\mathbf{x}^{(t)})}{\partial x_1} & \cdots & \frac{\partial f_1(\mathbf{x}^{(t)})}{\partial x_N} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_N(\mathbf{x}^{(t)})}{\partial x_1} & \cdots & \frac{\partial f_N(\mathbf{x}^{(t)})}{\partial x_N} \end{bmatrix}^{-1} \begin{bmatrix} f_1(\mathbf{x}^{(t)}) \\ \vdots \\ f_N(\mathbf{x}^{(t)}) \end{bmatrix}$$

- Updating rule for the Armington model:

$$(UR-4) \quad \mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \left[\nabla \mathbf{f}(\mathbf{w}^{(t)}) \right]^{-1} \mathbf{f}(\mathbf{w}^{(t)})$$

$\mathbf{w} \equiv [w_i]_{i=1}^N$ is the vector of wage rates, $f_i(\mathbf{w}) \equiv D_i(\mathbf{w}) - S_i(\mathbf{w})$ is excess labor demand function in country i , with $D_i(\cdot)$ and $S_i(\cdot)$ given by eq. (5).

Newton's method

- Root-finding algorithms that exploit partial derivatives extend significantly beyond the basic Newton's method which we covered, employing numerous advanced techniques and approaches:
 - e.g., calculate partial derivatives via numerical approximation instead of providing their exact formulas
 - See Chapter 3 of “Applied Computational Economics and Finance”, Miranda and Fackler
- In most programs such as Matlab, Julia, and Python, a range of pre-programmed solvers are available which you can use for your specific problem.
 - This is in contrast to Method #1 where we require to devise a distinct algorithm for each specific model.

Which method to use?

- In practice, there is no guarantee that either of the two approaches converge quickly, or even converge at all.
 - except in special cases where there is an analytical result.
- The tradeoff is that Newton's methods use the information on the shape of the obj function more efficiently but only at the cost of calculating the derivatives.
- Wise to learn both. If one does not work, you can switch to another.
 - Note: Newton's methods are of many types themselves. Similarly, one can devise model-based algorithms in multiple ways.
- Next, we'll show that the difference between the two approaches is, after all, smaller than it might initially seem.

Method #1 as an approximation of the Newton's method

- Consider the *relative* excess demand function,

$$f_i(\mathbf{w}) \equiv \frac{D_i(\mathbf{w})}{S_i(\mathbf{w})} - 1$$

where:

$$\frac{D_i(\mathbf{w})}{S_i(\mathbf{w})} = w_i^{-\sigma} \frac{a_i^{\sigma-1}}{L_i} \sum_{j=1}^N \frac{b_{ij} d_{ij}^{1-\sigma}}{\sum_{\ell=1}^N b_{\ell j} d_{\ell j}^{1-\sigma} (w_{\ell}/a_{\ell})^{1-\sigma}} w_j L_j$$

- Approximation 1.** Ignore off-diagonal elements of the inverse Jacobian matrix:

$$\frac{\partial f_i(\mathbf{w})}{\partial w_j} = \frac{\partial \left(\frac{D_i(\mathbf{w})}{S_i(\mathbf{w})} - 1 \right)}{\partial w_j} \approx 0, \quad j \neq i$$

- Under Approximation 1, no matrix calculation is needed:

$$w_i^{(t+1)} = w_i^{(t)} - \left[\frac{\partial f_i(\mathbf{w}^{(t)})}{\partial w_i} \right]^{-1} \times f_i(\mathbf{w}^{(t)})$$

Method #1 as an approximation of the Newton's method

- Reproduce the equilibrium condition,

$$\frac{D_i(\mathbf{w})}{S_i(\mathbf{w})} = w_i^{-\sigma} \times \sum_{j=1}^N \left[\frac{a_i^{\sigma-1}}{L_i} \frac{b_{ij} d_{ij}^{1-\sigma}}{P_j^{1-\sigma}} w_j L_j \right]; \quad P_j^{1-\sigma} = \sum_{\ell=1}^N \left[b_{\ell j} d_{\ell j}^{1-\sigma} \left(\frac{w_\ell}{a_\ell} \right)^{1-\sigma} \right]$$

- A change in w_i alters $D_i(\mathbf{w})/S_i(\mathbf{w})$
 - *directly* via PE forces that alter expenditure shares on origin i —through $w_i^{-\sigma}$;
 - *indirectly* via GE forces that alter income levels and price indices—through $\{w_j L_j\}_j$ and $\{P_j^{1-\sigma}\}_j$ in the summation term.
- **Approximation 2.** Ignore the indirect GE effects.

- Under Approximation 2,

$$\frac{\partial (D_i(\mathbf{w})/S_i(\mathbf{w}) - 1)}{\partial w_i} \approx -\sigma \frac{D_i/S_i}{w_i}$$

Method #1 as an approximation of the Newton's method

- Updating Rule (UR-4) under Approximations 1 and 2:

$$\begin{aligned}w_i^{(t+1)} &= w_i^{(t)} - \left[\frac{\partial f_i(\mathbf{w}^{(t)})}{\partial w_i} \right]^{-1} f_i(\mathbf{w}^{(t)}) \\&= w_i^{(t)} - \frac{w_i^{(t)}}{-\sigma D_i^{(t)} / S_i^{(t)}} \left(D_i^{(t)} / S_i^{(t)} - 1 \right)\end{aligned}$$

which can be written as:

$$\text{(UR-5)} \quad w_i^{(t+1)} = \left[1 + \frac{1}{\sigma} \frac{D_i^{(t)} / S_i^{(t)} - 1}{D_i^{(t)} / S_i^{(t)}} \right] \times w_i^{(t)}$$

- (UR-5) is a special case of (UR-3) in which $g(\cdot)$ is specified as:

$$g(x) = 1 + \frac{1}{\sigma} \frac{x-1}{x}, \quad x > 0$$

Method #1 as an approximation of the Newton's method

- Even more interestingly,

$$\begin{aligned}w_i^{(t+1)} &= \left[1 + \frac{1}{\sigma} \frac{D_i^{(t)}/S_i^{(t)} - 1}{D_i^{(t)}/S_i^{(t)}} \right] \times w_i^{(t)} \\&= \left(1 - \mu^{(t)} \right) w_i^{(t)} + \mu^{(t)} \left(D_i^{(t)}/S_i^{(t)} \right) w_i^{(t)}; \quad \mu^{(t)} = \left[\sigma D_i^{(t)}/S_i^{(t)} \right]^{-1}.\end{aligned}$$

That is, (UR-5) is equivalent to (UR-2) when the dampening parameter in iteration t is $\mu^{(t)} = \left[\sigma D_i^{(t)}/S_i^{(t)} \right]^{-1}$

- Intuitively, when demand is more elastic (higher σ), demand becomes more sensitive to any price change. So, a higher level of dampening is more likely to prevent iterations to oscillate or diverge.